Work Sampling and Ergonomics

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Work Sampling

- Introduced by L.H.C. Tippet in 1934
- Has been in common use since 1950
- Estimates the proportion of time that a worker spends working.
- We make a large number of random observations and note if the worker is working or idle.

Basic formulae

$$\bar{p} = \frac{x}{n} = \frac{\text{number observed in classification}}{\text{Total number of observations}}; \quad s_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Upper estimate of proportion$=\bar{p} + ks_p$
Lower estimate of proportion$=\bar{p} - ks_p$

Confidence level	**0.90**	**0.95**	**0.99**	**0.999**
k	**1.645**	**1.960**	**2.576**	**3.291**

How many observations are needed?

- Let us say $\bar{p}$=0.5

- At 95% confidence level, suppose error is ±0.01. (Value may lie between 0.49 and 0.51).
- Then

$$k\,s_p = 1.96\sqrt{\frac{\bar{p}\left(1-\bar{p}\right)}{n}} = 1.96\sqrt{\frac{0.5\left(1-0.5\right)}{n}} = 0.01$$

$$\Rightarrow n = 9604$$

How many observations are needed if % p?

- Let us say $\bar{p}$=0.5

- At 95% confidence level, suppose error is ±0.01. (Value may lie between 0.49 and 0.51).
- Then

$$k\,s_p = 1.96\sqrt{\frac{\bar{p}\left(1-\bar{p}\right)}{n}} = 1.96\sqrt{\frac{0.5\left(1-0.5\right)}{n}} = 0.01$$

$$\Rightarrow n = 9604$$

How many observations are needed if idle time is 10% and you estimate should be accurate enough to tell that idle time is 9 to 11%?

- Here $\bar{p}$=0.1
- At 95% confidence level, suppose error is $\pm$0.01. (Value may lie between 0.9 and 0.11).
- Then

$$k\,s_p = 1.96\sqrt{\frac{\bar{p}\left(1-\bar{p}\right)}{n}} = 1.96\sqrt{\frac{0.1\left(1-0.1\right)}{n}} = 0.01$$

$$\Rightarrow n = 3457.44 \approx 3458$$

- We should take higher value of round-off.

Performance standards

$$\text{Normal time} = \frac{\text{total time} \times \text{proportion of working} \times \text{performane rating}}{\text{Total number of pieces produced}}$$

$$\text{Standard time} = \text{Normal time} + \text{Allowances for delays, fatigue, and personal time}$$

Introduction to Ergonomics

From the point of view of Work Study

It is an applied science concerned with designing and arranging things people use so that the people and things interact most efficiently and safely

What is Ergonomics?

- - The study of designing equipment and systems for human well-being
- - Aims to improve efficiency and reduce discomfort
- - Applied in workplaces, industry, and daily life

Objectives of Ergonomics

- - Enhance efficiency and productivity
- - Reduce injury and discomfort
- - Improve overall well-being and safety

Branches of Ergonomics

- - Physical Ergonomics: Focuses on human anatomy and movement
- - Cognitive Ergonomics: Studies mental processes and decision-making
- - Organizational Ergonomics: Improves work policies and teamwork

Principles of Ergonomics

- - Fit the task to the worker
- - Maintain a neutral body posture
- - Reduce repetitive movements and strain
- - Ensure proper lighting and workstation setup

Ergonomics in Workplace Design

- - Proper desk and chair height
- - Monitor at eye level
- - Keyboard and mouse placement for wrist comfort
- - Importance of breaks and stretching

Common Ergonomic Issues

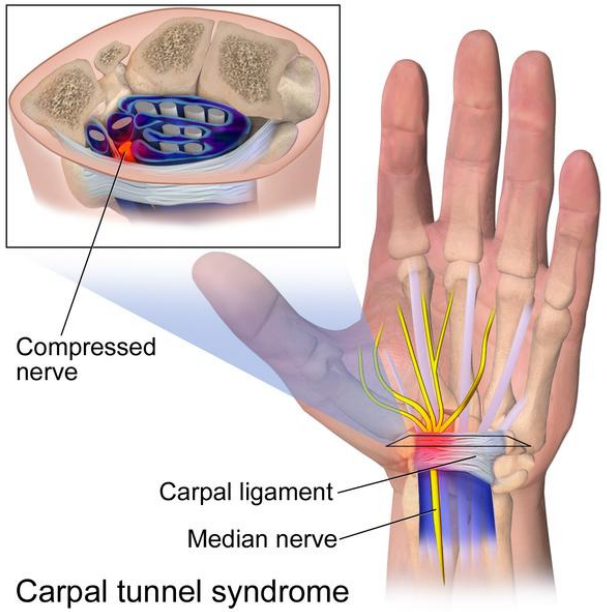
- - Musculoskeletal disorders (MSDs)

Musculoskeletal disorders (MSDs) are injuries or diseases that affect the muscles, bones, joints, tendons, nerves, and other tissues that support the body. MSDs can range from minor aches and pains to long-term conditions that lead to disability.

- - Carpal tunnel syndrome
- - Back and neck strain
- - Eye strain and fatigue

Carpal tunnel syndrome

Carpal tunnel syndrome (CTS) occurs when the median nerve in the wrist is compressed. This can lead to pain, numbness, tingling, weakness, or muscle damage in the hand and wrist



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Back and neck strain, often caused by overuse, poor posture, or injury, can lead to pain, stiffness, and limited movement, requiring rest, gentle stretching, and sometimes over-the-counter pain relief.

Eye strain and fatigue, often caused by prolonged screen time or focusing on tasks, can lead to symptoms like dry, itchy, or burning eyes, blurred vision, and headaches.

Solutions and Best Practices

- - Maintain correct posture while sitting/standing
- - Use ergonomic furniture
- - Take frequent breaks and stretch
- - Adjust screen brightness to reduce eye strain

Ergonomics in Manufacturing and Industry

- - Designing tools to reduce strain
- - Automating repetitive tasks
- - Using safety gear to prevent injuries

Ergonomics in Computer Use

- - Correct sitting posture
- - Monitor distance and height
- - Blue light filters to protect eyes
- - Avoid prolonged sitting

A **blue light filter** is a feature or tool designed to reduce the amount of blue light emitted by screens—such as those on smartphones, computers, and tablets.

Ergonomic Tools and Technologies

- - Adjustable desks and chairs
- - Wearable posture correctors
- - AI-based ergonomic assessments

Case Studies in Ergonomics

- - **Company**: Ford Motor Company
  Problem: Workers in assembly lines were experiencing repetitive strain injuries due to awkward postures and excessive reach.
  Solution:
- Ford implemented 3D motion capture and virtual simulation tools to analyze postures.
- Redesigned workstations to bring tools and parts within optimal reach.
- Introduced height-adjustable platforms.
  Outcome:
- 70% reduction in musculoskeletal disorder (MSD) risk factors.
- Improved assembly time and worker satisfaction.

Company: A multinational call center
Problem: High rates of neck, shoulder, and back pain among employees.
Solution:
•Replaced fixed desks with height-adjustable sit-stand desks.
•Provided ergonomic chairs and monitor stands.
•Introduced hourly stretch breaks.
Outcome:
•Reported 40% reduction in discomfort.
•Improved employee retention and call handling efficiency.

Company: Microsoft
Problem: Carpal tunnel syndrome and finger fatigue in typists.
Solution:
•Developed split keyboard design with a natural angle to reduce wrist strain.
•Added palm rests and feedback-sensitive keys.
Outcome:
•Widely adopted ergonomic standard across the tech industry.
•Positive feedback from long-term users with fewer complaints of RSI (Repetitive Strain Injury).

Ergonomics and Productivity

- - A well-designed workspace increases efficiency
- - Employees are healthier and more engaged
- - Less absenteeism due to work-related injuries

Myths and Misconceptions About Ergonomics

- - Ergonomics is only for offices (Myth: It applies to all industries)
- - Expensive furniture is always ergonomic (Myth: Proper setup is key)
- - Sitting is better than standing all day (Myth: Balance is necessary)


ESC
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
DEL
CTRL
WIN
ALT
FN

Future Trends in Ergonomics

- - AI-driven workplace adjustments
- - Smart wearable devices
- - Virtual ergonomics training

Conclusion and Key Takeaways

- - Ergonomics enhances efficiency and health
- - Proper posture and workstation setup are essential
- - Implement small changes for big benefits